

# Siddhartha Shakya

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## PROFILE SUMMARY

Backend Software Engineer with a First-Class Honors in Computer Science and hands-on experience designing and building scalable backend systems, REST APIs, and end-to-end applications. Proficient in Python, SQL, and cloud-based tooling with a proven ability to integrate third-party APIs, automate workflows, and ship production-ready code. Adept at translating real-world problems into working prototypes, collaborating with cross-functional teams, and applying strong fundamentals in data structures and system design.

## TECHNICAL SKILLS

|                             |                                                                                      |
|-----------------------------|--------------------------------------------------------------------------------------|
| <b>Languages</b>            | Python, SQL, C/C++                                                                   |
| <b>Backend &amp; APIs</b>   | REST APIs, third-party API integration (GEE, fintech), FastAPI, GitHub Actions CI/CD |
| <b>Databases</b>            | PostgreSQL, MySQL, MongoDB, DuckDB                                                   |
| <b>Cloud &amp; DevOps</b>   | Docker, Git, GitHub, YAML, Databricks, Snowflake                                     |
| <b>Data &amp; Analytics</b> | Apache Airflow, dbt, PySpark, Pandas, Streamlit, Jupyter Notebooks                   |

## WORK EXPERIENCE

**Backend & AI Engineer (Intern)** | *Mach24 Orbitals* **Oct 2025 – Feb 2026**

- Designed and built a scalable backend system integrating the Google Earth Engine (GEE) REST API to automate satellite imagery retrieval across custom areas of interest and time ranges, eliminating manual workflows entirely.
- Owned end-to-end data pipeline development: automated ingestion, quality assurance, cleaning, and feature engineering to deliver production-ready datasets for downstream analytics and model training.

**Graduate Teaching Assistant** | *Herald College Kathmandu* **Aug 2024 – Sep 2025**

- Designed and delivered Big Data curriculum covering databases (SQL, MongoDB), Python, and distributed systems (Hadoop, PySpark) to 120+ students, achieving a 15% improvement in assessment scores.
- Delivered High Performance Computing workshops on parallel programming and CUDA GPU computing, reinforcing strong fundamentals in system design, concurrency, and performance optimization.

## PROJECTS

**EduFin Investigation** | *Financial Risk Analysis* | [Link](#) **2026**

- Built and deployed scalable data pipelines on Databricks using PySpark and SQL to ingest and analyze a 15,000+ record loan portfolio, uncovering a Rs. 12 Crore financial discrepancy through data-driven risk segmentation.
- Translated a real-world financial problem into a working analytical prototype, applying clean, maintainable code practices and delivering actionable risk mitigation insights.
- **Tech Stack:** Python, Databricks, PySpark SQL, Delta Tables, REST APIs, Cloud Storage

**NYC Parking Violations** | *End-to-End Data Pipeline* | [Link](#) **2025**

- Built a multi-layer Medallion Architecture ELT pipeline transforming raw NYC Open Data into analytics-ready metric tables, demonstrating strong fundamentals in databases and system design.
- Automated production deployments via GitHub Actions CI/CD with separate dev/prod database profiles, shipping clean, maintainable, and production-ready code.
- **Tech Stack:** dbt Core, DuckDB, SQL, Python, GitHub Actions, YAML

## EDUCATION

**PG Diploma in Data Science** **Feb 2026 – Expected Mar 2027**

- *Manipal Academy of Higher Education*

**BSc (Hons) in Computer Science** **Feb 2022 – Jun 2024**

- *University of Wolverhampton*
- Honors: **First Class**

## CERTIFICATIONS

|                                                |           |                             |
|------------------------------------------------|-----------|-----------------------------|
| <b>AI Fluency: Framework &amp; Foundations</b> | Anthropic | <a href="#">Certificate</a> |
| <b>MLOps Bootcamp</b>                          | Udemy     | <a href="#">Certificate</a> |
| <b>SQL Bootcamp</b>                            | Udemy     | <a href="#">Certificate</a> |